Importing Library

In [1]:
```
import pandas as pd
import matplotlib as mp
import matplotlib.pyplot as plt
import scipy
from datetime import datetime, timedelta
import math
import numpy as np
```

Importing Data

Sanitizing data

In [2]:
```
df = pd.read_csv("data.csv")
df = df.dropna(axis=1, how="all") # drop all columns that only has NaN
df = df.dropna() # drop rows that has NaN
df.reset_index(inplace=True)
df["Date"] = df["Date"].astype(str)
df["S&P Comp P"] = df["S&P Comp P"].astype(float)
df["CPI"] = df["CPI"].astype(float)
df["Long Interest Rate GS10"] = df["Long Interest Rate GS10"].astype(float)
df["Real Total Return Price"] = df["Real Total Return Price"].str.replace(",
df["Real TR Scaled Earnings"] = df["Real TR Scaled Earnings"].str.replace(",
df["Excess CAPE Yield"] = df["Excess CAPE Yield"].str.rstrip('%').astype(flo
df["10 Year Annualized Stock Real Return"] = df["10 Year Annualized Stock Re
df["10 Year Annualized Bonds Real Return"] = df["10 Year Annualized Bonds Re
df["Real 10 Year Excess Annualized Returns"] = df["Real 10 Year Excess Annua
df.head()
```

Out[2]:

	index	Date	S&P Comp P	Dividend	Earnings	CPI	Date Fraction	Long Interest Rate GS10	Real Price	Rea Dividen
0	120	1881.01	6.19	0.27	0.49	9.42	1881.04	3.70	220.66	9.4
1	121	1881.02	6.17	0.27	0.48	9.51	1881.12	3.69	217.74	9.5
2	122	1881.03	6.24	0.28	0.48	9.51	1881.21	3.69	220.21	9.7
3	123	1881.04	6.22	0.28	0.47	9.61	1881.29	3.68	217.34	9.7
4	124	1881.05	6.50	0.29	0.47	9.51	1881.37	3.67	229.39	10.0

5 rows × 21 columns

Datetime

Fixing the date, currently the date in the data is in the format of **(yyyy.mm)** but does not give us the actual day. We will instead use the "Date Fraction" column as that will allow us to calculate a more precise date of the data (e.g. 1871.46 mean 46% of year 1871 has passed which is around 168 days from Jan 1, 1871; around mid-June).

In [3]:
```
def fraction_date(fd):
    yr = int(fd)
    year_frac = fd - yr

    start_date = datetime(yr, 1, 1)
    days = 366 if pd.Timestamp(year=yr, month=12, day=31).is_leap_year else
    days_passed = math.ceil(year_frac * days)

    result = start_date + timedelta(days=days_passed)
    return result

df["Date"] = df["Date Fraction"].apply(fraction_date)
df.drop("Date Fraction", inplace=True, axis=1) # Drop it, no longer needed
df.head()
```

Out[3]:

	index	Date	S&P Comp P	Dividend	Earnings	CPI	Long Interest Rate GS10	Real Price	Real Dividend	R To Ret Pr
0	120	1881-01-16	6.19	0.27	0.49	9.42	3.70	220.66	9.45	408
1	121	1881-02-14	6.17	0.27	0.48	9.51	3.69	217.74	9.53	404
2	122	1881-03-19	6.24	0.28	0.48	9.51	3.69	220.21	9.70	410
3	123	1881-04-17	6.22	0.28	0.47	9.61	3.68	217.34	9.78	406
4	124	1881-05-17	6.50	0.29	0.47	9.51	3.67	229.39	10.06	430

Data Exploration Part 1

For our first aspect of data exploration, we will look at if there is a correlation between the real total bond returns, and the real dividend returns over the time frame of our data set (1881-2016)

For this, we will use hypothesis testing. Based on background knowledge from some finance and economics classes, I believe that dividends and stocks are likely positively correlated during certain inflationary periods, and negatively correlated during deflationary periods, but over the long run (such as our data set) the correlation is likely non-existent. As such, I will use this as the Null Hypothesis

HO: Real Dividends and Real Total Bond Returns are not related in any way (have no correlation)

HA: Real Dividends and Real Total Bond Returns are related in some way (have some positive or negative correlation)

We will use the pearson correlation test to evaluate the null hypothesis

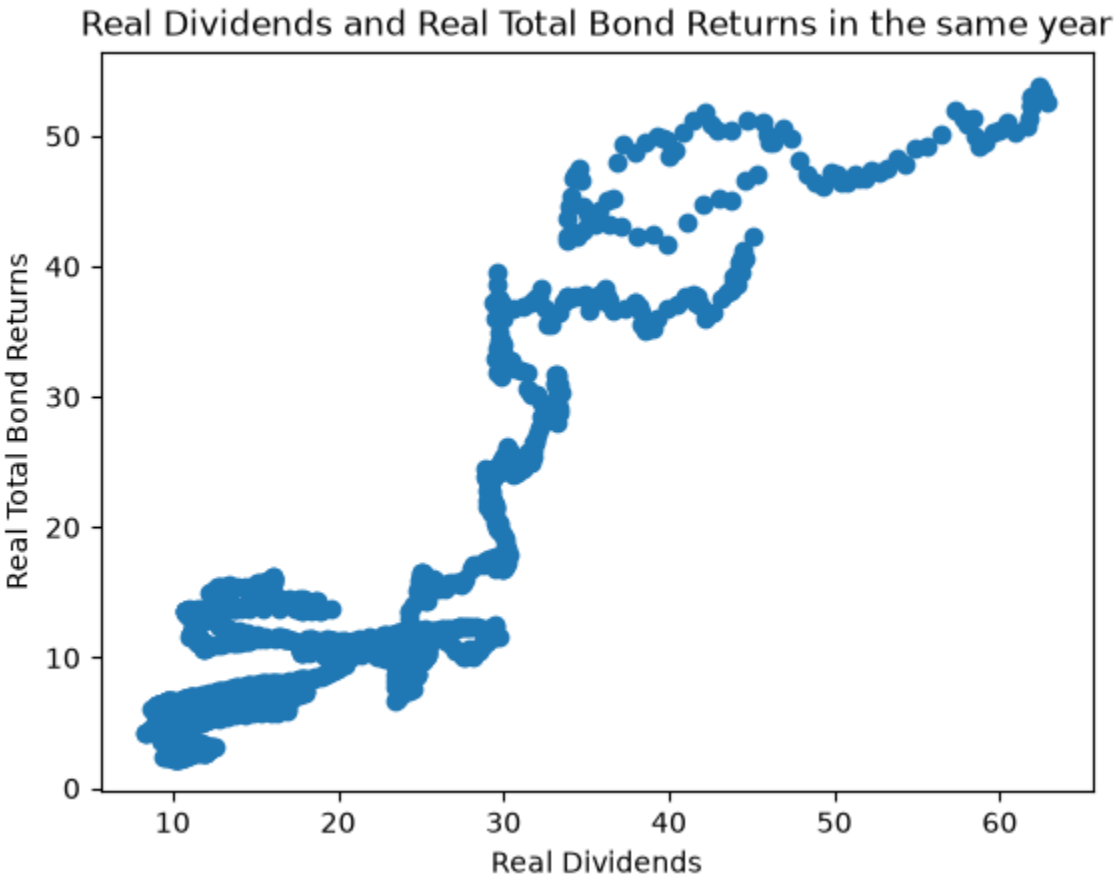
We will apply an alpha value of 0.05

In [4]:
```
fig, ax = plt.subplots()
ax.scatter(df["Real Dividend"],df["Real Total Bond Returns"])
plt.title("Real Dividends and Real Total Bond Returns in the same year")
plt.xlabel("Real Dividends")
plt.ylabel("Real Total Bond Returns")

s, p= scipy.stats.pearsonr(df["Real Dividend"],df["Real Total Bond Returns"]

print("P Value: ", p)
print("Correlation Statistic: ", s)
```

```
P Value:  0.0
Correlation Statistic:  0.8680507673810401
```



In this case, the P value is impressively 0, implying that the null hypothesis is very wrong and that over the course of the data, the dividend and return value are heavily correlated. As the correlation statistic "r" is positive, it implies a positive correlation. When one goes up, the other also goes up. It should be noted that this does not mean that increased dividends result in increased bond returns, or vice versa. Rather there is likely some confounding variable -- such as the economy -- that is affecting both,

and this variable may not be properly capture by our datasets. It's hard to gauge the concept of the economy in a simple variable.

It's worth noting that the P value is likely not truly zero, rather, it's just so low that the computer renders it as zero. This is likely due to the high correlation value across a rather large dataset causing the t-test statistic used by the pearsonr function (the t-test statistic is roughly 70.5) and thus results in an absurdly low p value once the t-statistic is analyzed.

Data Exploration Part 2

For our second data exploration, we'll examine whether the returns are normally distributed. Because the normal distribution is a common assumption in many statistical tests, determining whether our stock data is normally distributed will help us decide which tests to run going forward. It will also let us adjust our conclusions when using models that assume normality, reflecting that extreme events are more common than those models predict.

In addition, we will visualize the real total return with a Q-Q plot. A Q-Q plot will compare our data’s quantiles with the quantiles of a normal reference distribution, allowing us to see whether our data is normal.

HO: returns are normally distributed (skewness = 0, excess kurtosis = 0)

HA: returns are not normally distributed

We will use the Jarque-Bera hypothesis test to assess normality. We chose the Jarque-Bera test because it tests whether the data's skewness and kurtosis match those of a normal distribution. Kurtosis refers to the "tailedness" of a probability distribution. If the data has more extreme outliers than a normal distribution, it will have higher kurtosis. Because returns typically depart from normality in two ways, asymmetry (skewness) and fat tails (kurtosis), we chose the Jarque-Bera test to assess the normality of the financial data. Our alpha value will be 0.05.

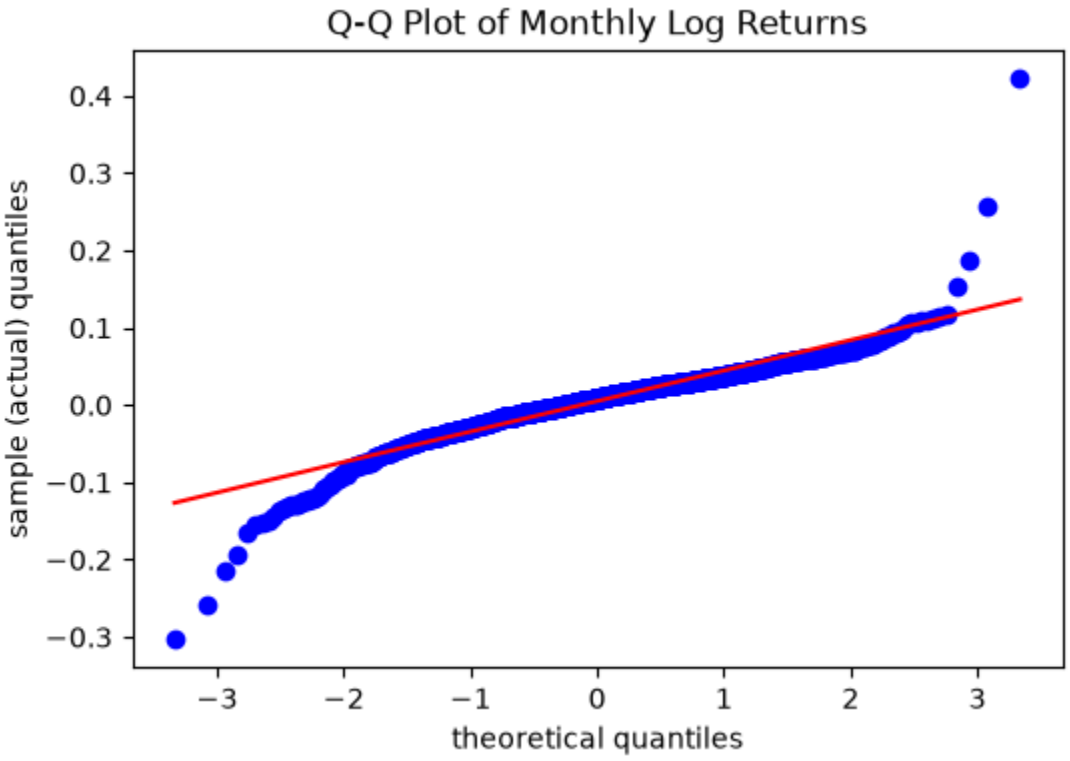
In [5]:
```
tr = df["Real Total Return Price"].dropna()
#takes the log of every return price and subtracts it against the previous
#thus it returns the change for each month
r = np.log(tr).diff().dropna()
#bc real Total Return Price is already adjusted for inflation and includes r
#r is a real total return: what an investor actually gained in purchasing po

print("Jarque–Bera Hypothesis testing")
jb_stat, jb_p = scipy.stats.jarque_bera(r)
print(f"JB = {jb_stat:.1f} P= {jb_p} log10(p) ≈ {-jb_stat / 2 / np.log(10):.
print("skew:", scipy.stats.skew(r), "| excess kurtosis:", scipy.stats.kurtos


plt.figure(figsize=(6, 4))
# dist='norm' checks against normal distribution
scipy.stats.probplot(r, dist="norm", plot=plt)

plt.title("Q-Q Plot of Monthly Log Returns")
plt.xlabel("theoretical quantiles")
# what we would expect if the data was actually normal
plt.ylabel("sample (actual) quantiles")
# actual returns from r sorted smallest to largest
plt.show()
```

```
Jarque–Bera Hypothesis testing
JB = 8664.6 P= 0.0 log10(p) ≈ -1881.5
skew: -0.26872994093830915 | excess kurtosis: 11.285667708038627
```



In this case, the P value returned 0, so we took log10 of the P value to estimate it and find the true P value; it is roughly 10^-1881.5, far smaller than our alpha of 0.05. Because the p-value provides strong evidence against the null hypothesis, we reject it.

The data appear normally distributed on the Q-Q plot, but the tails show substantial kurtosis. Based on this, we conclude that although the bulk of the distribution is approximately normal, departures are concentrated in the tails. One possible explanation for the extreme departure might be the many economic events our data spans, like the great depression. A good next step would be to analyze whether these departures are linked to economic events. Because our dataset spans major historical events like WWII and the Great Depression, we can compare distributions throughout eras and test how much kurtosis is attributable to the most extreme observations.

Data Exploration Part 3: Comparing 10-Year Stock and Bond Returns

For our third data exploration, we will compare the historical 10-year annualized real returns of stocks and bonds. We will group the returns into different percentage ranges and use a bar chart to visualize how frequently each occurs. This analysis will help us understand which investment has historically provided higher returns and how their return distributions differ.

In [6]:
```
# Return categories
bins = [-np.inf, 0, 0.02, 0.05, 0.08, 0.12, np.inf]
labels = ["Negative", "0–2%", "2–5%", "5–8%", "8–12%", "Above 12%"]

# Categorize stock and bond returns
stock_categories = pd.cut(
    df["10 Year Annualized Stock Real Return"],
    bins=bins,
    labels=labels
)

bond_categories = pd.cut(
    df["10 Year Annualized Bonds Real Return"],
    bins=bins,
    labels=labels
)

# Count observations in each category
stock_counts = stock_categories.value_counts().reindex(labels)
bond_counts = bond_categories.value_counts().reindex(labels)

# Combine results
comparison = pd.DataFrame({
    "Stocks": stock_counts,
    "Bonds": bond_counts
})

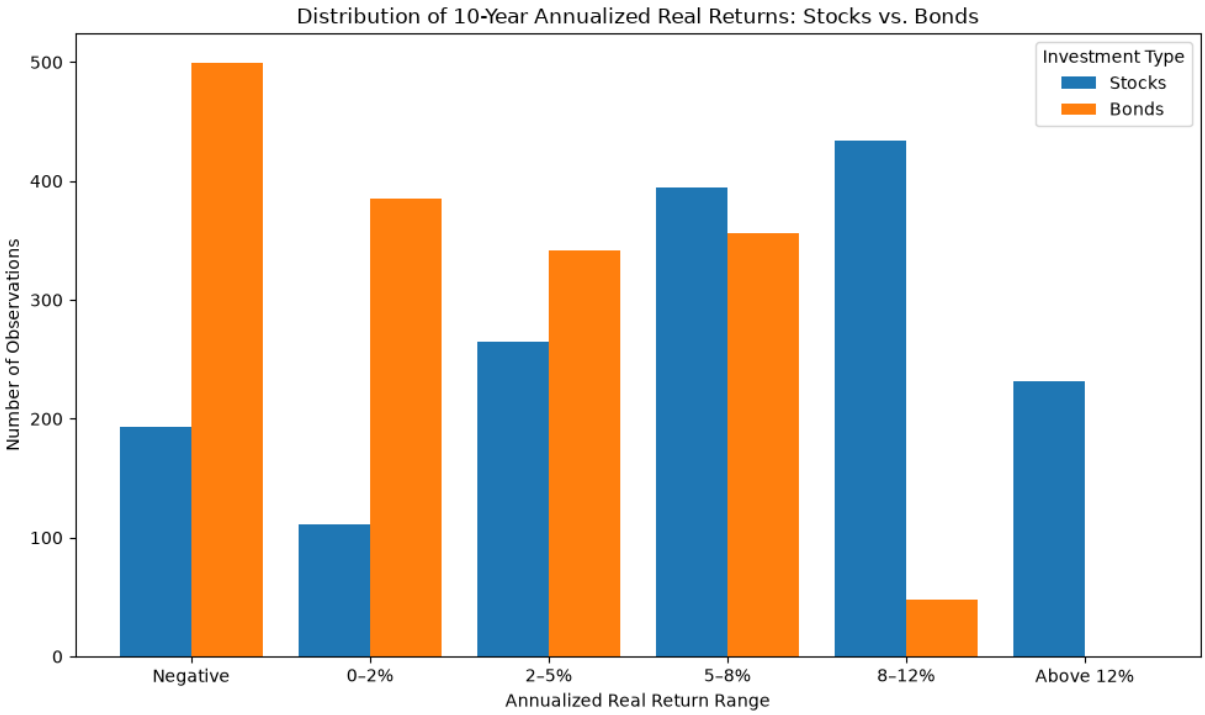
# Create grouped bar chart
ax = comparison.plot(kind="bar", figsize=(10, 6), width=0.8)

ax.set_title("Distribution of 10-Year Annualized Real Returns: Stocks vs. Bo
ax.set_xlabel("Annualized Real Return Range")
```

```
ax.set_ylabel("Number of Observations")
ax.legend(title="Investment Type")

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Display category counts
display(comparison)
```



	Stocks	Bonds
Negative	193	499
0–2%	111	385
2–5%	265	342
5–8%	395	356
8–12%	434	48
Above 12%	232	0

Based on our results, stocks generally had higher 10-year annualized real returns than bonds. Looking at the graph, stocks had 666 observations with returns above 8%, while bonds only had 48. Bonds also had more negative returns, with 499 observations compared to 193 for stocks. However, stocks had a higher standard deviation of 5.14% compared to 3.40% for bonds, meaning their returns varied more. Overall, our analysis shows that stocks tended to perform better over the long term, but their returns were also less consistent than bonds.